

Abstract

European green crabs (*Carcinus maenas*) are one of the most prolific invasive species in the marine environment. They have spread all over the world from their native ranges off the shores of eastern Europe and have recently been introduced to the waters of the Salish Sea in the Pacific Northwest region of the United States. These crabs pose a significant threat to the environment through destructive behaviors, wide diets, and low vulnerability to environmental stressors, all of which allow for them to outcompete other native crab species like *Hemigrapsus spp.* Current work is focused on removal efforts of European green crabs in the Salish Sea to decrease the amount of recruitment to the area, but modern trapping methods are also known to catch native crabs and fish. Alternative trapping method known as the “crab slab” was tested to determine if it provides a lower stress environment for native crabs in the presence of a European green crab. Results show that there is no significant difference in stress of native crabs in different trap types or with a European green crab. This shows that native crabs are not experiencing decreased stress environments in “crab slab” traps when compared to other current trap models. Better understanding of how native crabs respond to European green crab removal efforts is important to make sure that anthropogenic impacts to the environment are minimized, and removal efforts are not exhibiting excess stressors on native species.

Introduction

European green crabs (*Carcinus maenas*) pose a significant threat to non-native environments (Behrens Yamada & Gillespie, 2008). They are one of the most destructive species of invaders across the entire planet (Ens et al., 2022). Many different factors and adaptations allow the crabs to be strong and destructive invaders. They have a generalist diet with a large range of target prey, many of which overlap with prey of native crabs, leading to high levels of competition (Klassen & Locke, n.d.). They also sometimes will consume native crabs themselves, directly decreasing population sizes (Jensen et al., 2002). European green crabs have a high tolerance to environmental change like temperature or salinity, which allows them to expand into larger geographical ranges and survive swings in conditions better than native counterparts (Leignel et al., 2014; Tepolt & Somero, 2014). They also are known to dig up and damage critical habitats like eelgrass beds from burrowing behaviors (Behrens Yamada & Gillespie, 2008; Cosham et al., 2016). All these factors combined make European green crabs one of the top 100 worst invasive species ever introduced to new habitats (Simberloff & Rejmanek, 2019).

European green crabs, in their native range, live along the coast of the eastern Atlantic Ocean from Norway all the way down to Morocco (Fisheries, 2026). The crabs first invaded the waters of North America sometime in the 1800s from the shores of Europe in the ballast waters of shipping and merchant vessels (Ens et al., 2022). Others believe that they traveled around hiding among shellfish aquaculture shipments or materials (Ens et al., 2022). In the 1990s the crabs spread to the west coast of North America, starting in San Francisco and moving northward due to currents carrying larvae around (“European Green Crab,” 2022). No matter the mode of transportation, they have since taken over the eastern North American shoreline, the western North American shoreline, the southwestern tip of South Africa, and the southeastern tip of Australia (Yamada et al., 2021). The potential range of the species extends to a much larger area, leading to increasing concerns about slowing down the spread of the species and increasing

invasion prevention measures (Behrens Yamada & Gillespie, 2008; Dunagan, n.d.; Tepolt & Somero, 2014).

One of the places that has been recently exposed to European green crabs are the waters of the Pacific Northwest and the Salish Sea (Yamada et al., 2021). European green crabs were first confirmed in this area in 2012 in Sooke Basin, British Columbia and on the San Juan Islands in 2016 (“European Green Crab,” 2022). Two species of crabs that are competitors and threatened by the presence of European green crabs in the Pacific Northwest are *Hemigrapsus oregonensis* and *Hemigrapsus nudus* (deRivera et al., 2011; Jensen et al., 2002). These crabs live in a similar range to where European green crabs are moving into and exist in the same ecological niche (Jensen et al., 2002). European green crabs sometimes will even predate on native *Hemigrapsus* species (Klassen & Locke, n.d.). Therefore, further invasion of European green crabs into the Salish sea would pose a threat to the populations of *Hemigrapsus spp.*

Not only do European green crabs cause a multitude of environmental issues, they also can cause significant damage to the success of local fisheries and affect the economy (Walton et al., 2002). Increasing European green crab populations in this area would increase damage to eelgrass beds and estuary habitats (critical to salmon recovery efforts), as these crabs are known to burrow and eat shellfish populations that hold these habitats together (Ens et al., 2022). They may disrupt aquaculture processes, salmon fishery recovery efforts, and Dungeness crab fisheries, which combined are worth an estimated 1.79 billion dollars annually (*Coastal Commercial Dungeness Crab Fishery* | *Washington Department of Fish & Wildlife*, n.d.; Fisheries, 2024; Mcmillan, n.d.; Walton et al., 2002). Overall, there is not only a very high environmental demand to study invasion prevention methods in the Pacific Northwest waters, but also an incredibly high economic risk to the state of Washington’s fisheries if European green crabs were able to fully invade the area.

There are currently many teams working to remove European green crabs from the Salish sea (Dunagan, n.d.; “European Green Crab,” 2022). Traditional crab traps are made of wire and are baited to lure crabs inside (Cauvel, 2024). The problem is that more than just European green crabs get caught inside. Many other native crab species or fish may also wind up trapped with the European green crab, and in such proximity with no chance to escape, chance of predation and exhibited stress may increase (Cauvel, 2024; Dunagan, n.d.). One group working to solve this problem is the Tulalip Tribe with their innovative design called the “crab slab” (Cauvel, 2024). This trap mimics prime habitat for European green crabs to entice them to come to the trap without the need for bait. Additionally, since the traps stay open to the environment, and native crab that is threatened from a European green crab will be able to just leave the trap and find a different location, while the European green crab stays inside the “crab slab” (Cauvel, 2024). To determine if the “crab slab” induces a lower stress response in native crab species experiments were conducted in a lab setting. This study analyzed a simulated “crab slab” environment and a simulated normal crab trap (ie. fukui trap or minnow trap) with or without the presence of a green crab in the water and measured multiple stress response variables. We hypothesized that the classic traps would induce a higher stress response in *Hemigrapsus spp.* and that the presence of European green crabs will induce a higher stress response in *Hemigrapsus spp.* This study will aid in the capture and removal efforts of European green crabs in the Pacific Northwest by better understanding how these efforts are affecting native competitors and will help increase the effectiveness of current European green crab removal efforts by understanding the stress impacts on the native ecosystem.

Methods

Crab traps were simulated using four smaller jars within 2 larger tanks. Two of the small jars were covered in blackout tape and filled $\frac{1}{3}$ of the way with small gravel collected from Golden Gardens beach to simulate the “crab trap” (see Figure 1). The other two small jars were lined with flexible mesh to simulate the fukui trap (see Figure 1). One of each small jar type was placed into the larger tanks, and a European green crab was placed within one of the large tanks (see Figure 2). *Hemigrapsus spp.* crabs were selected to all be around the same size (average 3.4 grams per crab) and all males. Once crabs were selected, they were labeled with ID tags. Then three *Hemigrapsus spp.* were randomly placed within each simulated trap for a total of 12 crabs across all treatment groups.



Figure 2. Simulated “crab slab” with black out tape on the outside and gravel located on the left. Simulated fukui trap with mesh lining the inside on the right.

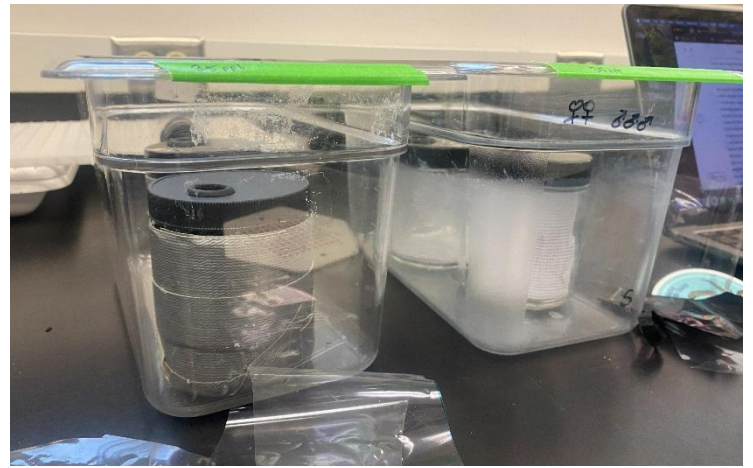


Figure 1. Simulated crab traps placed within each European green crab treatment tank. Tank on the left contained a European green crab for the duration of the experiment while the tank on the right did not.

Every week for three weeks crabs were removed from their respective traps, and righting time test was performed by flipping crab on its back and timing how long it took to right itself back over again. Righting times were averaged by treatment and analyzed with a Kruskal-Wallis for significance. Then 13 wells were set up filled with 13 mL of pre-mixed resazurin as according to the open-source protocol published by the University of Washington’s Roberts Lab (Huffmyer et al., 2025). 200 μ L samples of the resazurin for each well were collected before crabs were added. All 12 crabs were then placed in their wells for an hour, one staying empty as the control, and two more 200 μ L samples were collected with the crabs at 30 minutes and 60 minutes. Once the resazurin test has been completed crabs were rinsed with salt water and replaced back into their respective traps for another week. Resazurin fluorescence was normalized to starting values and crab mass and then analyzed with a Kruskal-Wallis test for significance. The final week of the experiment both righting time and respiration tests were complete before moving onto hemolymph extractions. Hemolymph was extracted from the joint connecting the legs to the body for every crab and then analyzed for the total protein counts. Analysis was conducted using a Kruskal-Wallis test for significance.

Results

Initial results show that there is no significant relationship in relative crab stress level and the type of trap they exist in. The first variable tested was righting time, which is used as a proxy of health in crabs (see Figure 3) (Schroeder-Spain et al., 2018). Results show no significant differences in righting time between treatment groups ($p = 0.47$). Analysis was run a second time after the removal of a significant outlier of 20.37 seconds (due to loss of limbs), and analysis was run again with no change of significance ($p = 0.26$). The second variable tested was respiration, which was used as a proxy for metabolic rate of the crab (see Figure 4). Results show no significant differences in respiration rate between treatment groups ($p = 0.781$). The third variable tested was total protein levels in the hemolymph, which was used as a proxy for energy stores utilized (see Figure 5). Results show no significant differences in protein levels across treatment groups. Two arrays were run at a full saturation level and a diluted level. Neither method produced significant differences ($p = 0.838$).

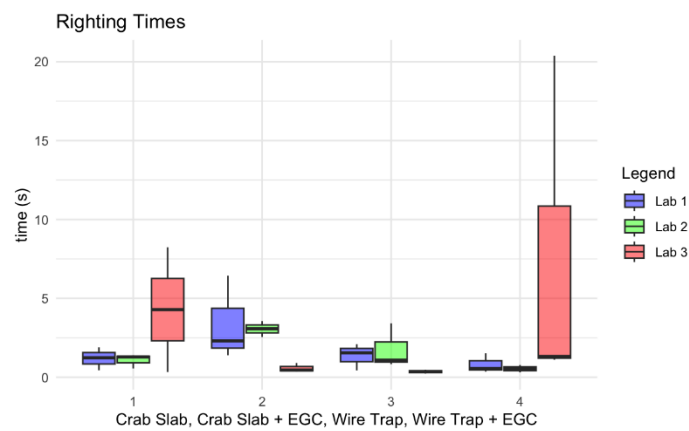


Figure 4. Righting time of each crab by treatment type across 3 weeks. Colors represent each treatment group. Largest increases in righting time after the three weeks were groups 1 and 4. Groups 2 and 3 experienced decreases over the experimental window. $p = 0.47$

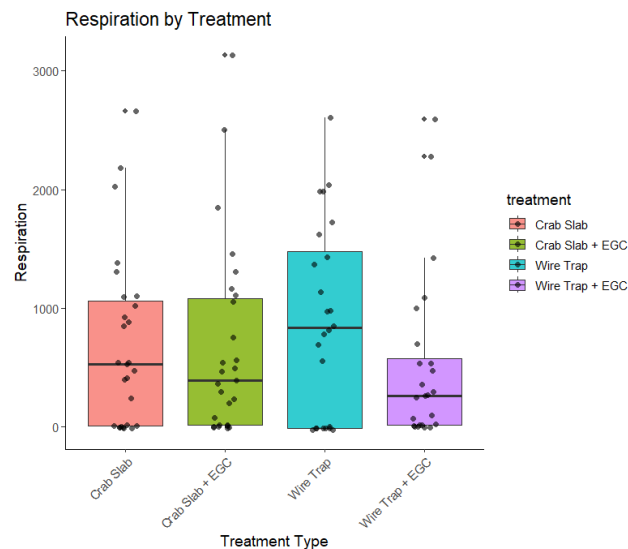


Figure 3. Respiration level of each crab group by treatment type grouped over the three weeks of the experiment. Colors represent each treatment group, and the points represent a single crab's respiration at one time point during the respiration test. Crab slab treatments have very similar respiration to one another compared to the wire traps, but no significant relationship was detected ($p = 0.781$).

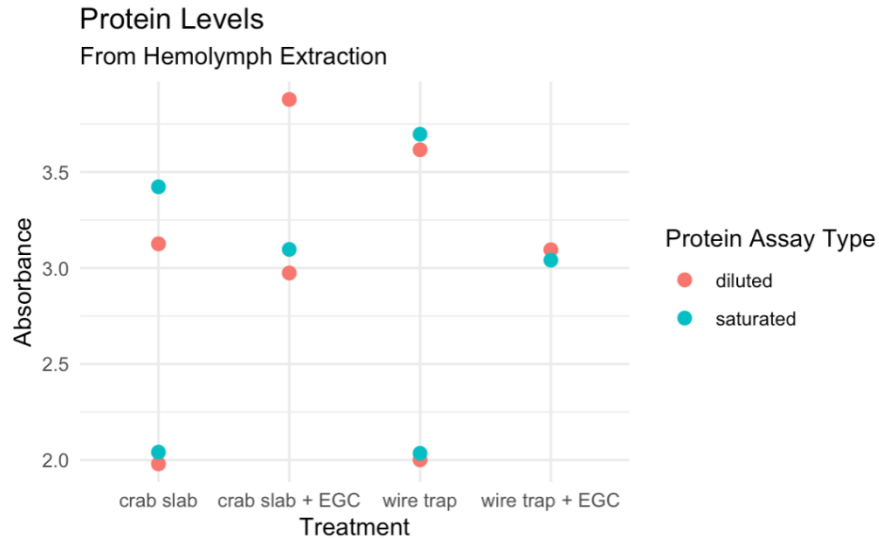


Figure 5. Total protein content of crabs with successful hemolymph extraction by treatment group. The two types of protein assay depicted across color. Results show no significant difference across treatments ($p = 0.838$)

Discussion

Overall results show that there is no difference in the stress induced by different crab traps or by the presence of European green crabs. Although we expected that the simulated fukui trap would be more stressful for the *Hemigrapsus spp.* there was no significance across the treatment groups in any of the variables tested. One consideration to these results is that this experiment used simulated crab traps that were much smaller than those used in the field to catch wild European green crabs. Future experiments should be run to see if the significance of the experiments differs if the full-sized traps were used.

These results will help inform current efforts to remove European green crabs from the Pacific Northwest by providing information to decrease the effects of these efforts on the natural environment. European green crab removal efforts that cause increasing amounts of damage or stress to native populations will decrease the overall effectiveness of the removal in the first place by causing an additional layer of harm. “Crab slab” traps, which are directly designed to mitigate these effects, are therefore more likely to be better for native crab species.

Resources

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